

FlowDesk – Product Requirement Document (PRD)

Version: 1.0 **Date:** May 2026 **Prepared By:** Agha Ali **Internship Assignment:** AI-Driven Product Development Program – Financia Solutions

1. Product Overview

Product Name: FlowDesk **Type:** SaaS-Based Task & Workflow Management System **Target Users:** Small to Medium Businesses, Startups, Remote Teams, Freelancers

What is FlowDesk?

FlowDesk is a cloud-based SaaS platform that helps teams manage their tasks, projects, and workflows efficiently in one place. Managers can assign tasks, track progress, set deadlines, and collaborate in real-time through a clean modern dashboard.

Problem It Solves

- Teams lose track of tasks assigned via WhatsApp or email
- No visibility on who is doing what
- No deadline tracking or accountability
- No centralized place to manage work

Solution

FlowDesk provides a centralized platform where tasks are created, assigned, tracked, and completed with full visibility for everyone.

2. Frontend Functionality

Public Pages

- Landing Page
- Login Page
- Register Page
- Forgot Password Page

Dashboard

- Summary cards: Total Tasks, Pending, In Progress, Completed
- Recent activity feed

- My Tasks section
- Upcoming deadlines widget
- Team performance overview

Task Management Module

- Create task (title, description, assignee, priority, deadline)
- List View and Kanban Board View
- Filter by status, priority, assignee, due date
- Edit, Delete, Complete tasks
- Add comments and attach files

Project Module

- Create projects and group tasks under them
- Project progress bar
- Project timeline view

Team Module

- View all team members
- Invite members via email
- Assign roles

Notifications

- In-app and email notifications

Reports Module

- Task completion report
- Team performance report
- Export as PDF or Excel

3. Backend Functionality

- User Authentication (Register, Login, Logout, Password Reset)
- JWT Token-based session management
- Task CRUD operations
- Project CRUD operations
- Role-based access control (RBAC)
- Email notification service
- File upload and storage
- Activity logging

- Search and filtering
 - Dashboard analytics
 - Report generation
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4. User Management

- Super Admin – Full system access
 - Admin – Manages one organization
 - Manager – Creates projects, assigns tasks, views reports
 - Member – Views and updates only assigned tasks
 - Guest – Read-only access
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5. Role & Permission Management

- Create Project: Admin, Manager only
 - Assign Tasks: Admin, Manager only
 - Update Own Tasks: Admin, Manager, Member
 - View Reports: Admin, Manager only
 - Manage Users: Admin only
 - Delete Tasks: Admin, Manager only
 - View Dashboard: Everyone
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6. Database Structure

Users Table: id, name, email, password_hash, role, created_at

Projects Table: id, name, description, owner_id, status, created_at

Tasks Table: id, title, description, project_id, assigned_to, created_by, priority, status, due_date, created_at

Comments Table: id, task_id, user_id, content, created_at

Notifications Table: id, user_id, message, is_read, created_at

7. APIs

Auth: POST /api/auth/register, /api/auth/login, /api/auth/logout, /api/auth/reset-password

Tasks: GET, POST /api/tasks | GET, PUT, DELETE /api/tasks/:id

Projects: GET, POST /api/projects | GET, PUT, DELETE /api/projects/:id

Users: GET /api/users | PUT, DELETE /api/users/:id

8. Security Considerations

- Passwords hashed using bcrypt
 - JWT Authentication with 24 hour expiry
 - HTTPS only (SSL/TLS)
 - Role-Based Access Control
 - Input validation to prevent SQL Injection and XSS
 - API rate limiting
 - Daily database backups
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9. Mobile App Workflows

Screens: Splash, Login, Dashboard, Task List, Task Detail, Create Task, Notifications, Profile

Mobile Features:

- Push notifications
 - Offline mode
 - Swipe to complete task
 - Mobile responsive UI
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10. AI Tools Used

- Claude AI – PRD writing and product thinking
 - Bolt.new – Frontend prototype generation
 - v0 by Vercel – UI component design
 - ChatGPT – Brainstorming features
 - Figma AI – UI/UX wireframing
 - GitHub Copilot – Code assistance
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11. Tech Stack

- Frontend: React.js / Next.js
 - Backend: Python (FastAPI or Django REST)
 - Database: PostgreSQL
 - Authentication: JWT + bcrypt
 - Hosting: Vercel (Frontend), Render (Backend)
 - File Storage: Cloudinary
 - Email: SendGrid
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